

Mg(OH)₂ LAYER COATING ON THE SURFACE OF CADMIUM ELECTRODES AND ITS EFFECT ON Ni-Cd CELL PERFORMANCE

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Abstract

Deposition of Mg(OH)₂ on to the bare Cd electrode surface was done by dipping bare Cd electrode in magnesium nitrate and KOH solutions. The Cd electrode was first immersed in the magnesium nitrate solutions of different concentrations and dried in the open atmosphere and then immersed in the 10% KOH solution for five minutes and dried in the open atmosphere. The effect of Mg(OH)₂ deposit on the Ni electrode surface was electrochemically studied by electrochemical impedance spectroscopy, cyclic voltammetry and chronoamperometry. A “H” type cell was used to construct the Ni-Cd cell and to study the effect of Mg(OH)₂ deposit on Cd electrode surface. The experimental results revealed that addition of the smaller quantities (~5%) of Mg(OH)₂ does not affect the electrochemical characteristics of cadmium electrode.

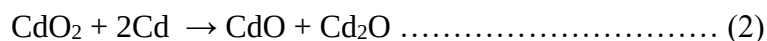
Key words

Battery; Electrodes; Surface Modification; Cyclic Voltammetry; EIS.

1.Introduction

In alkaline storage batteries, alkaline substances like NaOH, KOH are used as electrolytes [1]. In 1881, Fellx de Lahande and Georges Chaperson patented a cell, which could be called the first alkaline cell. It was a primary cell. In this, the active materials copper oxide as positive electrode and zinc plate as negative were used and electrolyte was KOH or NaOH. The emf of the cell was 0.88V. Later system was developed in several ways. In 1900 Edison substituted cadmium for zinc with the intention of obtaining a reversible cell by elimination of soluble zinc materials [2].

In 1901 Waldmer Junger found out that Cd-Fe, Cd-Cd systems, were very valuable, as they gave an energy up to 22.5 W/Kg [3], the chemical reactions for the above systems as proposed by Junger were,



In 1910 Junger after some modification of the above system, prepared cells by using black Nickel hydroxide as positive electrode. Cadmium and Iron sponge as negative electrode that was very successful [4]. Since the advent of first Nickel cadmium cell by Waldmer Junger, technology has developed well owing to inherent advantages of the system. Relatively cheap in comparison with Ag electrode, Cd-Cd design found favors in varied application such as from large capacity cells for stand by application to the effect by scalable cylindrical cells [5]. The system was pliable and many researchers studied the systems intensively and contributed to the development of technology [6-10]. In Nickel cadmium cell the positive electrode is made by structured Nickel oxy hydroxide active material. This active material coated are precipitated in the grid depending upon the type of electrodes. At the negative electrode during discharge the cadmium is converted into Cd^{+2} ion. In the Cd-Cd cells, the metallic cadmium represents the charged condition and is discharged but is precipitated as cadmium hydroxide and accounts for its low solubility in alkaline solution. Self-discharge is one of the factors which reduces the reaction or capacity of batteries. Self-discharge in cell happens in many ways due to the accumulation of discharged products. In Nickel cadmium batteries Nitrate ions present in electrode coming from electrode manufacturing process plays an important role. It reduces the performance of positive electrode and gives reduced output. The present study aims to deposit magnesium hydroxide on the Cd electrode of Cd-Cd cell and study the cell performance.

2.Experimental

For the experimental study and for evaluating charge performance of the Nickel electrode, three different concentrations of Magnesium Nitrate solutions were prepared, namely 5%, 10% & 15%. The Cd and Cd electrodes were fabricated by using electrochemical impregnation process are soaked in these solutions and performances of these electrodes were evaluated.

The preparation of additives in varying concentration, processing methods, evaluation are detailed as follows.

2.1 Preparation of an active Cadmium hydroxide :

It is an electro chemical impregnation technique in which the electrode is subjected to electrical polarization in a Magnesium Nitrate solution. Counter electrode is inert which is used as positive electrode. When the impregnation is over the electrode is treated with hydroxide.

2.2 Preparation of Magnesium Nitrate solutions :

A 5% $\text{Mg}(\text{NO}_3)_2$ solution has been prepared by dissolving in DM (Demineralized) water and made up to 1000 ml. Also 10% and 15% Magnesium Nitrate solutions were prepared, by dissolving 100g and 150g Magnesium Nitrate respectively in DM water and solutions are made up to 1000ml.

2.3 Preparation of 10% KOH solution

100 g of KOH pellet was dissolved in DM water (Demineralized) and made up to 1000ml. This solution is stored in a one-liter beaker.

2.4 Soaking the electrodes in 5% $\text{Mg}(\text{NO}_3)_2$ solution :

13 cm X 7 cm of six Cd electrodes were taken. One liter of Magnesium Nitrate solution was taken in a glass beaker. Electrodes were then soaked in the Magnesium Nitrate solution, and they were kept for 10 minutes in the solution. The Cd electrodes that were soaked, were taken and dried at ambient temperature. The dried electrodes were then soaked in 10%KOH solution again for 5 minutes.

The same evaluation method for determining Magnesium Nitrate solution retention in electrodes was followed for the solutions prepared with 10% and 15% concentration levels. The same technique was followed and the $\text{Mg}(\text{NO}_3)_2$ retention in the electrodes was evaluated.

2.5 Conversion of Magnesium Nitrate to Magnesium hydroxide at the electrode's surface:

The electrodes were soaked in the 10% KOH solutions for 10 minutes, dried at ambient condition. Then the dried electrodes were again soaked in 10% KOH solution for 5 minutes. The soaked electrodes were then dried. The same procedure is followed for the electrodes in 10 % and 15 % $\text{Mg}(\text{NO}_3)_2$ solutions respectively.

2.6 The electrode washing

The soaked electrodes were washed for 5 minutes in running DM water. Now, the washed electrode was taken and washed again in a beaker till the pH of washed water is neutral. If hydroxide ions are not properly washed carbonate formation will take place and removing this carbonate would be a difficult task. After washing, the electrodes were dried at 60°C for 1 hour.

2.7 Electrochemical Impedance Spectroscopy

The Ni-Cd cell was formed using H type glass cell containing 6M KOH solution. The Cd electrode served as working electrode, while Cd electrode served as counter electrode. Hg/HgO electrode was used as reference electrode. The electrochemical impedance spectra were recorded by superimposing a sinusoidal voltage of 10 mV amplitude at open circuit potential and also at different applied DC potentials after recording cyclic voltammograms.

2.8 Cyclic Voltammetry

The cyclic voltammograms were also recorded for bare and $\text{Mg}(\text{OH})_2$ deposited cadmium electrodes at the scan rates 20, 50 and 100 mV/S using the same cell set up employed for recording electrochemical impedance spectroscopy.

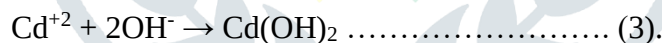
2.9 Chronoamperometry

Chronoamperometric studies have been carried out at different anodic potential using the same H type cell for evaluating the nature of $\text{Mg}(\text{OH})_2$ coatings over cadmium electrode.

3.RESULTS AND DISCUSSION

3.1 Cyclic voltametric analysis of Cd electrode surface

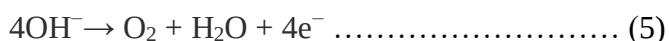
The cyclic voltammograms were first recorded for cadmium electrodes- plain as well as $\text{Mg}(\text{OH})_2$ deposited using Zahner Zennium XC electrochemical workstation at various scan rates ranging from 10mV/s to 100mV/s. Cyclic voltammograms of plain cadmium electrode are presented in the Figure.1a to 1d. The peak observed nearly at -1.25 V corresponds to adsorption of hydroxide ion onto the Cd electrode surface. The current observed for this peak is maximum for bare Cd electrode and with increase in amount of $\text{Mg}(\text{OH})_2$ deposit on the Cd electrode surface the current for adsorption of hydroxide ion increased for the $\text{Mg}(\text{OH})_2$ deposit from 5% $\text{Mg}(\text{NO}_3)_2$ solution to 10% $\text{Mg}(\text{NO}_3)_2$ solution and then decreased. This indicated the increase in surface coverage by $\text{Mg}(\text{OH})_2$ on Cd surface when the concentration of Nickel Nitrate solution is increased from 5% to 15% for surface modification. The second anodic peak observed nearly at 0.3 V indicated the oxidation of Cd to Cd^{+2} and the third peak at 0.4 V formation of $\text{Cd}(\text{OH})_2$, a passive layer formation, indicated by the chemical reaction,



The third anodic peak observed nearly at 0.13 V indicates the formation of Cd from $\text{Cd}(\text{OH})_2$ as follows,



Beyond this potential the oxygen evolution reaction, given in equation 5, takes place and is indicated a sharp increase in the current. The cathodic peak observed at the potential 0.13 V is the reversible peak of reaction 1, the reversible reaction is represented by equation 2. Beyond -1.2 V the reduction of O_2 to hydroxide ion takes place, which is represented by the reaction 3. The potential of oxygen reduction or hydroxide formation is not shifted with increase in surface coverage by $\text{Mg}(\text{OH})_2$. It can also be seen from the Fig.1a-1d that the peaks for oxidation of Cd and formation of $\text{Cd}(\text{OH})_2$ merged with increase in scan rate.



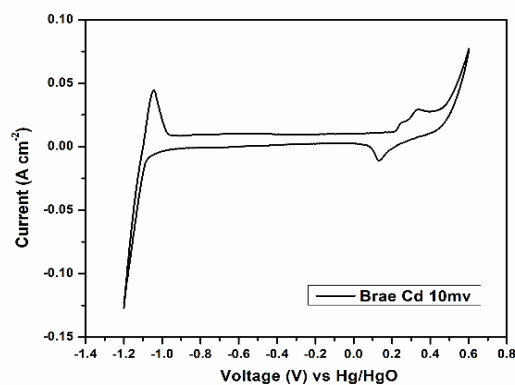


Figure.1a

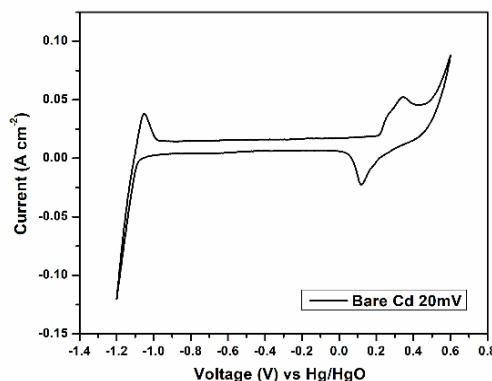


Figure.1b

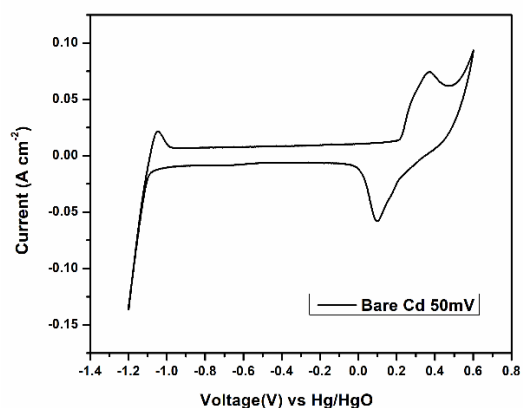


Figure.1c

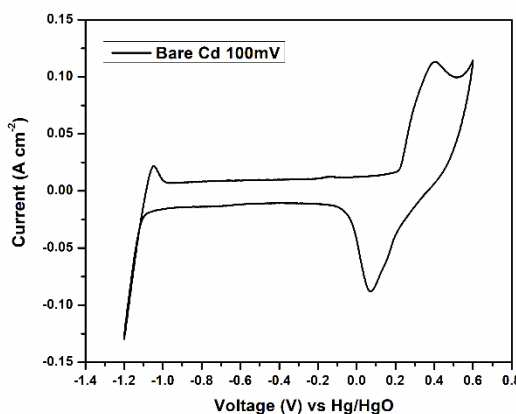


Figure.1d

Figure.1 Cyclic voltammograms of bare (1a) and $\text{Mg}(\text{OH})_2$ coated Cd electrode surface using 5% (1b), 10% (1c) and 15% (1d) $\text{Mg}(\text{NO}_3)_2$ solutions.

The anodic and cathodic peaks observed at different potentials for the bare and $\text{Mg}(\text{OH})_2$ coated Cd surface are presented in Figure.2. From these cyclic voltammograms, it can be seen that Cd surface coated with the $\text{Mg}(\text{OH})_2$ using 5% magnesium nitrate solution doesn't show an appreciable deviation from bare Cd surface and at high concentrations of $\text{Mg}(\text{OH})_2$ the electrochemical activity of Cd surface is appreciably affected.

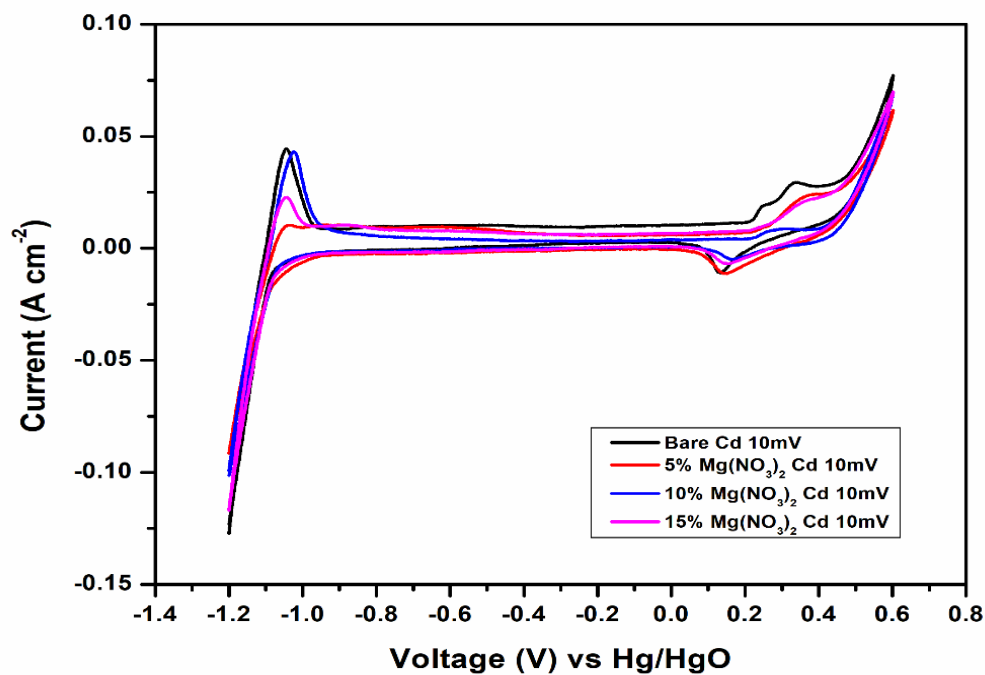


Figure.2 Effect of $\text{Mg}(\text{OH})_2$ coating on the cathodic and anodic peaks.

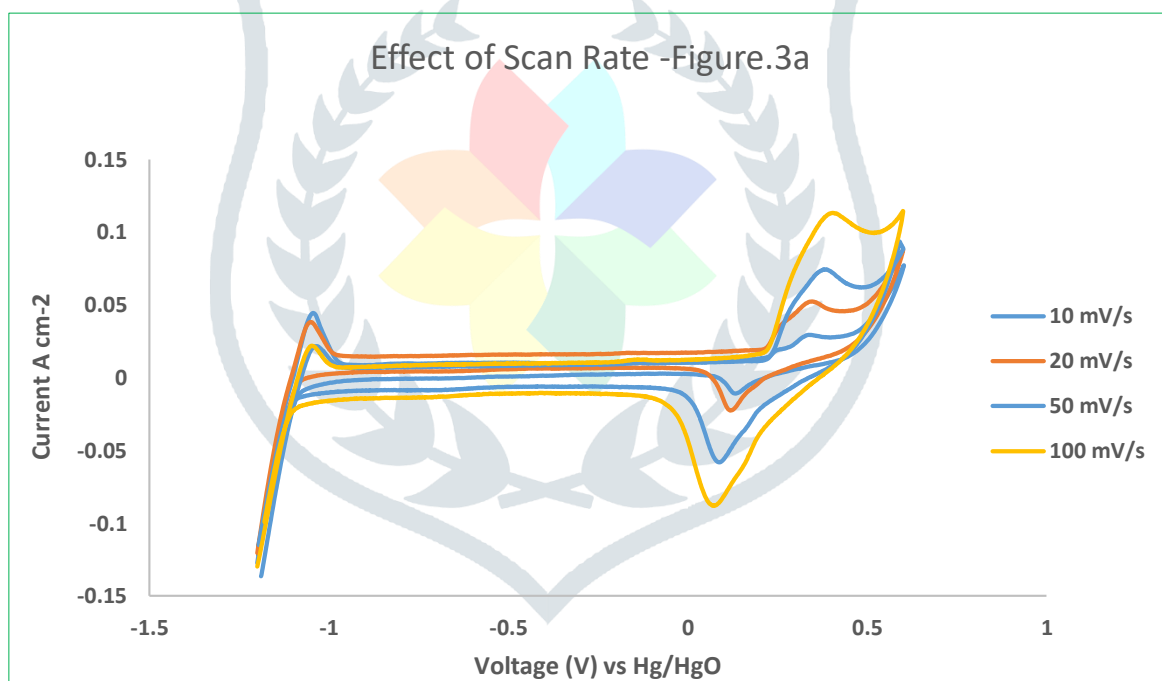


Fig.3a

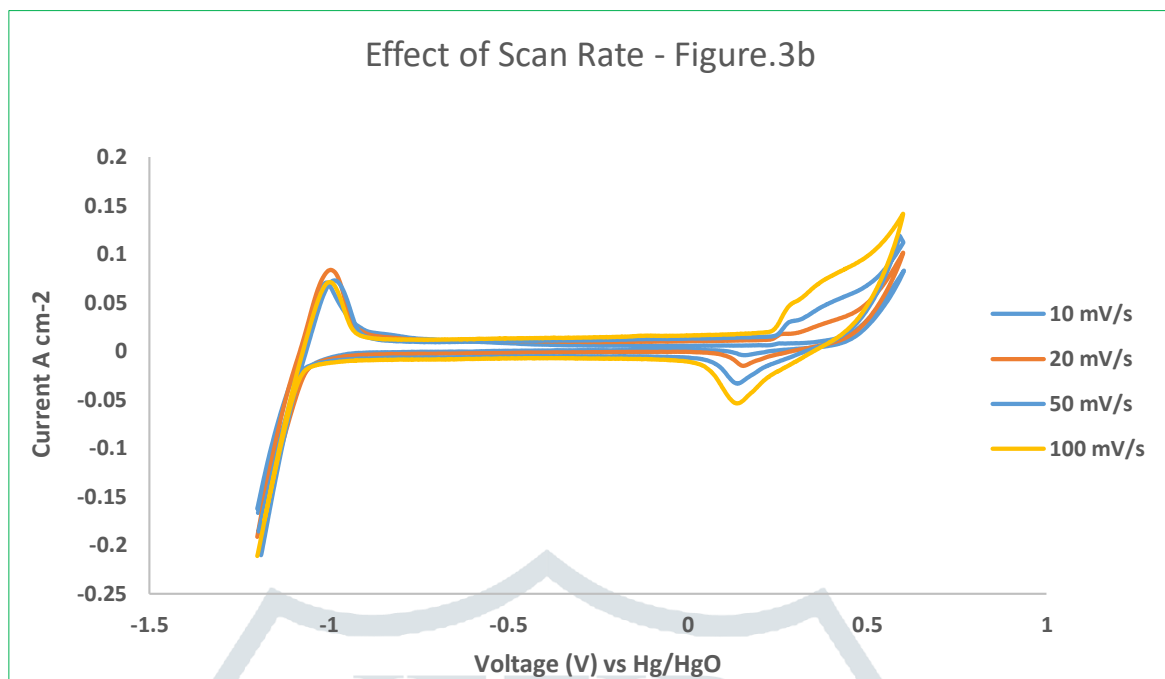


Figure.3 Effect of scan rate on the oxidation/reduction behavior of bare (3a) and $\text{Mg}(\text{OH})_2$ coated Cd

Cyclic voltammograms were also recorded to study the effect of scan rate on the oxidation/reduction behavior of bare and $\text{Mg}(\text{OH})_2$ coated Cd electrodes at different scan rates which are shown in Fig.3a and 3b. The experimental curves indicated that the peak current increased with sweep rate. Moreover, the peak potentials for oxidation and reduction are shifted to active and noble potentials respectively and the peak potential for adsorption of hydroxide ion doesn't change. The shift in peak potentials increased slightly when the Cd surface is coated with $\text{Mg}(\text{OH})_2$.

3.2 Electrochemical Impedance Spectroscopy of Bare and $\text{Mg}(\text{OH})_2$ coated Cd electrode

The Nyquist plots recorded for the bare and $\text{Mg}(\text{OH})_2$ coated Cd electrode are presented in Fig.4. From these plots it can be seen that the charge transfer resistance values decreased when the amount of $\text{Mg}(\text{OH})_2$ on Cd surface increases. This indicated the enhanced conductance at electrode solution interface with the amount of $\text{Mg}(\text{OH})_2$ precipitate on the Cd electrode surface. The experimental plots obtained fit well with the equivalent circuit shown in Fig.5. The R_u represents uncompensated solution resistance, R_p is charge transfer resistance, W_d is Warburg impedance which represents mass transport or diffusion at the electrode solution interface. Y_0 alpha represents a component of constant phase element which arises due to adsorption of anions like hydroxide ions onto the Cd electrode surface. The Nyquist plots conformed the filmed porous behavior of Cd electrode over wide frequency domain under charge transfer and diffusion control.

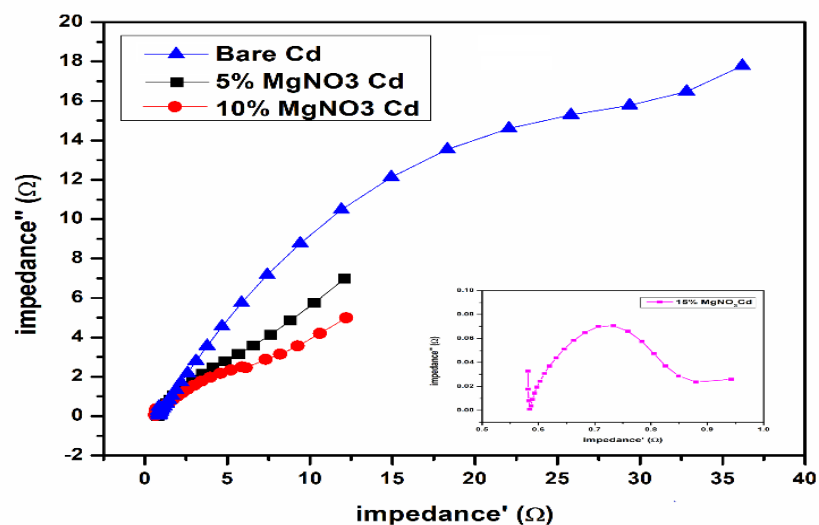


Figure.4 Nyquist plots for Bare and Mg(OH)₂ coated Cd electrodes in 6.0 M KOH solution

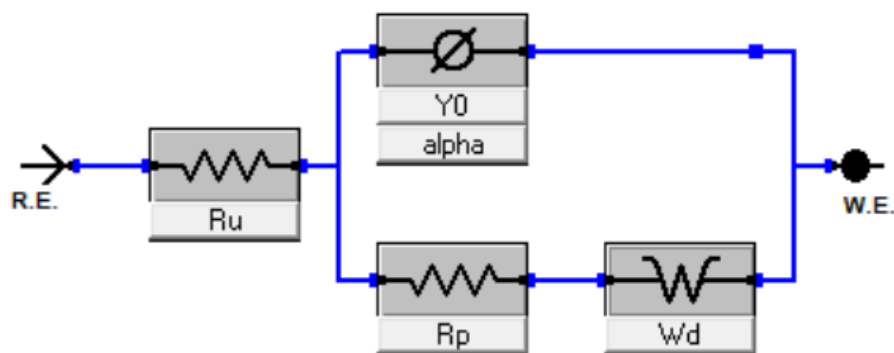


Figure.5 Equivalent Circuit fitting with the experimental Nyquist plots

3.3 Chronoamperometry

A series of experiments were carried out to study the anodic oxidation of Cd electrode in the alkaline solution in the absence and presence of Mg(OH)₂ precipitate. The current vs time curves so obtained are presented in Fig.6. Before recording chronoamperograms, firstly, the electrode was allowed to equilibrate in alkaline solution for 25 minutes until a steady open circuit potential is reached. An anodic voltage of -950 mV against Hg/HgO and stepped anodically towards -800 mV. The rapidly rising current response can be seen in the Fig.6 for stepping the potential of Cd electrode from -950 mV against Hg/HgO to -800 mV. This lies active dissolution region of Cd.

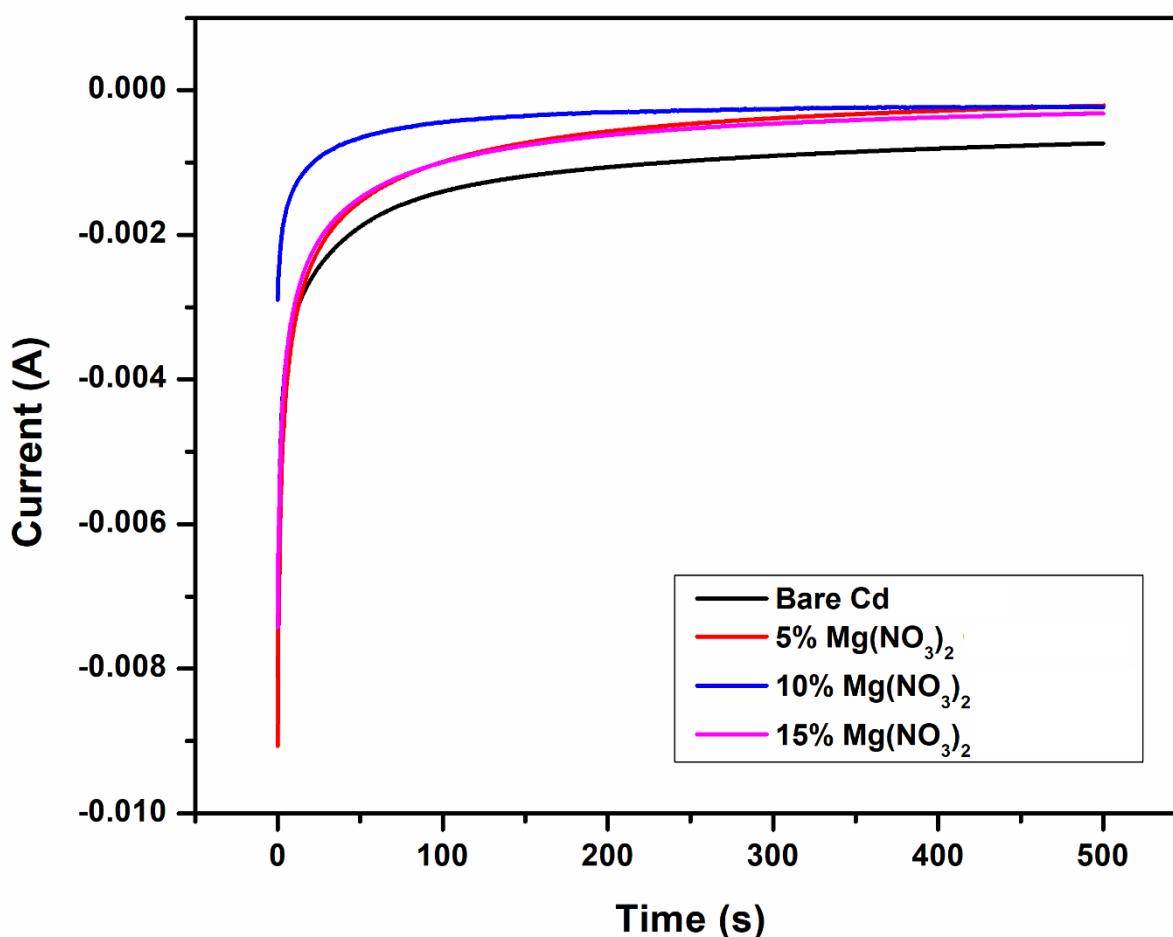


Figure.6 Current- time transients for Cd electrode in alkaline medium.

4.Conclusion

The following conclusions arrived in the present study.

$\text{Mg}(\text{OH})_2$ can be deposited on the surface of Cd coated with electrochemically active $\text{Cd}(\text{OH})_2$ and its electrochemical response can be studied by cyclic voltammetry, electrochemical impedance spectroscopy and chronoamperometry. The experimental results indicated that addition of the smaller quantities (~5%) of $\text{Mg}(\text{OH})_2$ does not affect the electrochemical characteristics of Nickel electrode and addition of the larger quantities (10 and 15%) affects the function of cadmium electrodes considerably.

References

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